EC306: Time Series Assignment

Word count: 1198 (excluding captions)

Question 1a:

Visual examination and an ADF test (Model B, no trend) confirmed stationarity of log unemployment, aside from a COVID-induced spike. To prevent this from biasing persistence, log unemployment was regressed on a COVID dummy and monthly seasons: residuals were extracted a cleaned series.

AR models up to 6 lags were estimated, where AIC/BIC were used for selection and Ljung-Box tests for admissibility. MA models were rejected as their residuals were not white noise. AR models below 3 lags produced autocorrelated residuals, with AIC selecting AR(6).

Forecasts can be generated dynamically, where one-step forecasts use the observed data and subsequent steps use earlier predicted values to iterate multiple steps ahead.

Question 1b:

Re-estimating an AR(6) on the forecast period revealed only 2 significant lags, motivating an AR(2). Model performance was weak overall (RMSE = 0.29; MAE = 0.136), with RMSE close to the series standard deviation, implying limited superiority to a naïve mean forecast.

However, Figure 1b-1 shows concentrated errors during COVID, as the pre-COVID period RMSE is 0.07. Additionally, the MAE being around 47% of the standard deviation supports that typical forecast errors are modest.

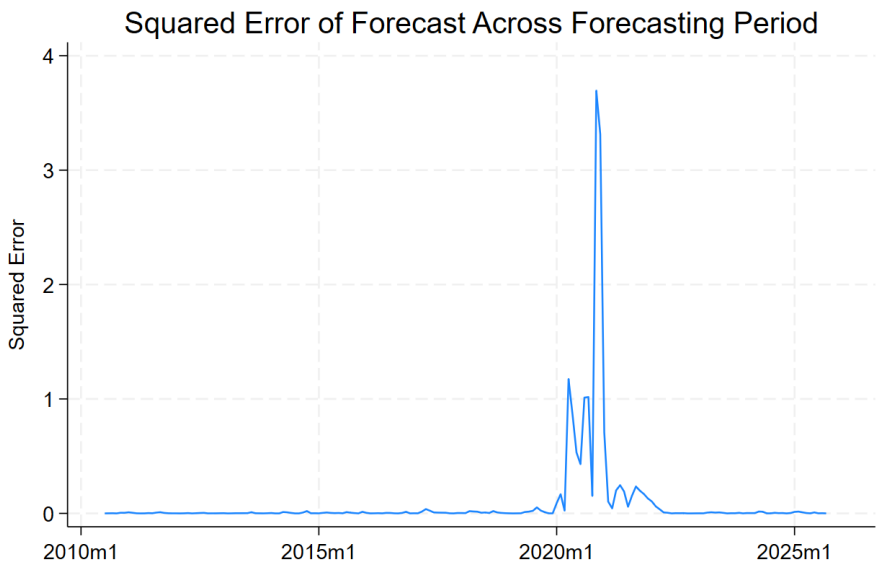

Figure 1b-1: Squared errors of forecast over forecasting period

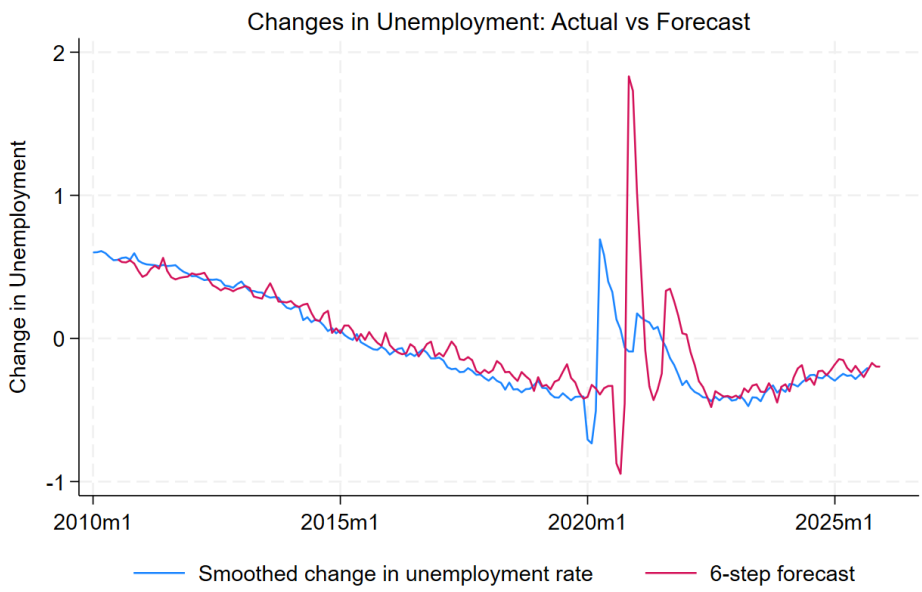

Figure 1b-2: Actual vs forecast changes in unemployment rates

Question 1c:

My January forecast (multivariate AR(1)) predicted 4.5% unemployment; the univariate AR(2) predicted 4.22%, which was closer to the real rate of 4.3% in March 2026.

The improved accuracy reflects a more thorough methodology, as the AR(1) model relied only on ACF/PACF plots, and levels data. My current approach applies the full Box-Jenkins method: systematically searching for the optimal lag order, screening model admissibility though Ljung-Box tests, correcting for outliers, and forecasting in differences.

My January model attempted to use the Federal Funds Effective Rate, aiming to use leading indicators to improve precision, but risked parameter uncertainty and overfitting, potentially worsening out-of-sample performance. The AR(2) model avoids this at the cost of ignoring useful external information.

Question 2:

To update forecasts as the quarter progressed, I augmented the baseline model of differenced GDP (an ARIMA(1,0,1)) with 2 monthly indicators (TCU and core CPI), which were averaged over however many months of the quarter had been observed at the time of forecasting. Each model is a regression of the differenced GDP on its own lag and the within-quarter averages for TCU and core CPI, which were re-estimated recursively. TCU is a leading indicator of output and can proxy supply-side productive activity, whilst core CPI captures underlying demand pressures that may co-move with GDP.

Figure 2-1 shows that within-quarter information was useful: the RMSE falls as the quarter progresses. Whilst this signals that it did not introduce much noise, RMSE differences are small. Additionally, the use of contemporaneous monthly values reflects how the RMSE improvements are an upper bound of feasible improvements in a real forecasting environment, as publication lags can result in that information not being immediately available at the time of forecasting.

Model	RMSE
Baseline ARIMA(1,0,1)	0.00420
1-month TCU and core CPI	0.00414
2-month TCU and core CPI	0.00407
3-month TCU and core CPI	0.00403

Figure 2-1: RMSE table for all models

Question 3a:

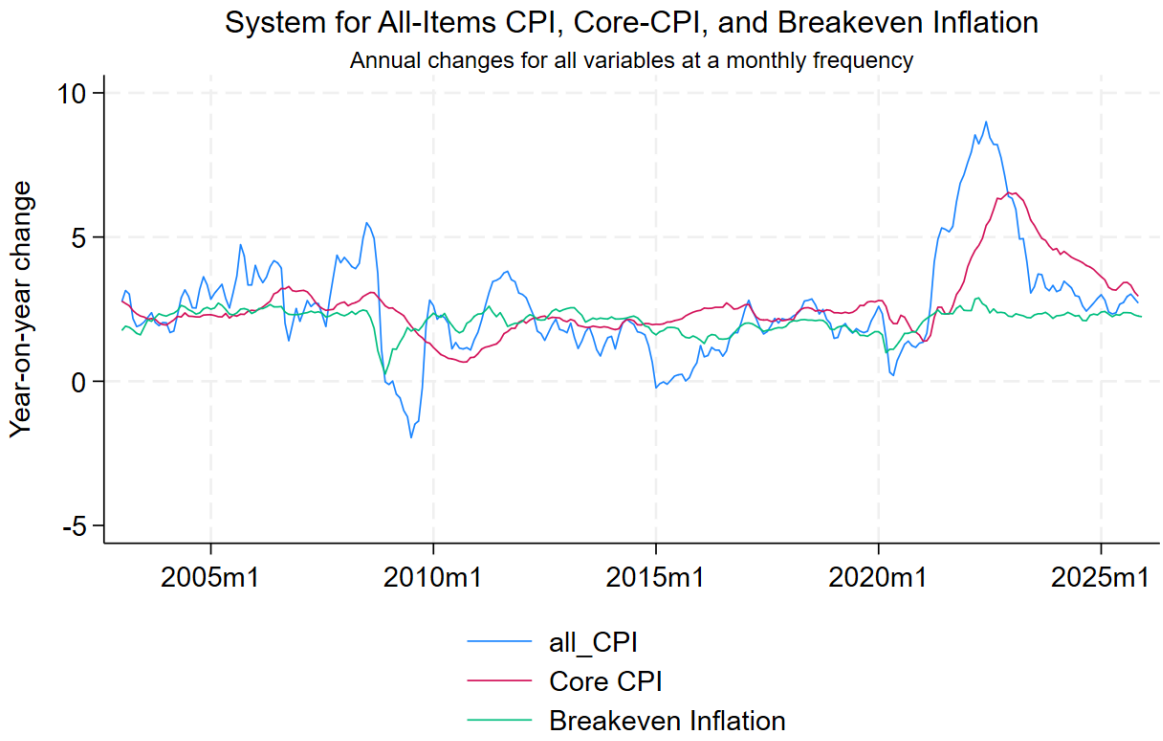

Figure 3a: System for all-items CPI, Core CPI, and Breakeven Inflation

Question 3b:

Variable	Mean	Standard deviation	Minimum	Maximum
πb_t (Breakeven inflation)	2.10	0.39	0.25	2.89
π_t (All-goods CPI)	3.52	2.87	-2.98	14.59
πc_t (Core CPI)	4.31	2.65	0.66	15.77

Table 3b: Summary of breakeven inflation, all-goods CPI, and core CPI

Figure 3a and Table 3b show that πt (all-goods CPI) is the most volatile. This is because it includes changes in food and energy prices, which are its main source of volatility as their prices are driven by temporary supply-side shocks rather than underlying economic trends. πct (core CPI) is much less volatile by removing this noise more accurately reflecting persistent inflationary trends.

The Federal Reserve's inflation-targeting policies are reflected in the characteristics of πbt, which represent consumer inflation expectations. Its mean of 2%, accompanied by its low standard deviation and stationarity, indicates how consumer expectations of inflation are anchored at 2% and are unlikely to deviate by a substantial amount due to the Federal Reserve's accumulated credibility.

ADF tests reject non-stationarity for all three series at the 1% level, testing Model B with drift.

Question 3ci:

Equation	Excluded	Testing	p-value
all-goods CPI	core CPI	Whether core CPI is Granger-causal to all-goods CPI	0.021
all-goods CPI	breakeven inflation	Whether breakeven inflation is Granger-causal to all-goods CPI	0.003
core CPI	all-goods CPI	Whether all-goods CPI is Granger-causal to core CPI	0.000
core CPI	breakeven inflation	Whether breakeven inflation is Granger-causal to core CPI	0.278
core CPI	ALL	Whether all variables are jointly Granger-causal to core CPI	0.000
breakeven inflation	all-goods CPI	Whether all-goods CPI is Granger-causal to breakeven inflation	0.033
breakeven inflation	core CPI	Whether core CPI is Granger-causal to breakeven inflation	0.532
breakeven inflation	ALL	Whether all variables are jointly Granger-causal to breakeven inflation	0.139

Figure 3ci-1: Granger causality Wald tests

Figure 3ci-1 indicates that πct and πt exhibit bi-directional Granger causality, which provides some evidence that πct is more forward looking than πt, but also that there is a feedback loop: persistent headline shocks eventually pass through to core prices as inflation becomes more embedded in the economy.

Figure 3ci-2 shows that a shock in πct rapidly increases πt, with the series reverting back to its mean relatively quickly. In contrast, a shock in πt increases pressures in πct more gradually, where the series is slower to revert back to its mean. The slower response supports the forward-looking argument proposed for πct, making it a more durable signal of future inflation.

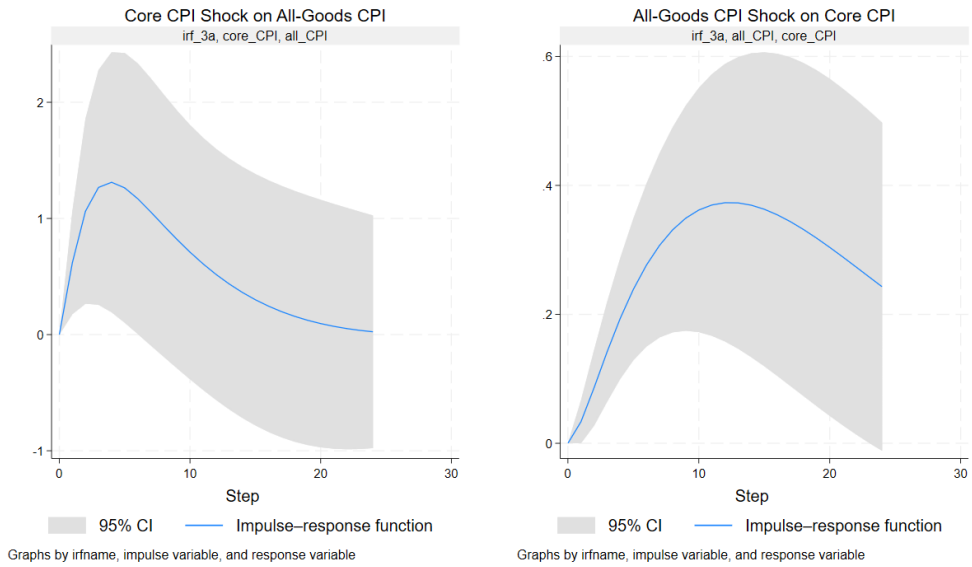

Figure 3ci-2: Impulse response functions for Core CPI and All-goods CPI

Question 3cii:

Figure 3ci-1 shows that πbt also exhibits bidirectional Granger-causality with πt and has no predictive power with πct. The feedback loop between πt and πbt can be explained by how πt is more reflective of consumers' experiences with prices on a daily basis: as such, changes to πt will cause them to update their expectations and change their spending patterns. In turn, this change will affect πt. The lack of predictive power between πbt and πct can be explained by how πbt captures total inflation expectations in financial markets, which may be driven by volatile changes in necessities like food and energy, rather than underlying trend inflation.

Question 4a:

Figure 4a-1 shows that over the next 12 months, a 1p.p. shock in πbt will cause a 1.81 p.p. increase in πt, peaking at 1.99pp after 8 months. Forecasts for πt will gradually keep increasing beyond the mean of πt , and πt will only begin reverting back to its mean after 8 months, where the effects of the shock will almost fully dissipate after 50 months.

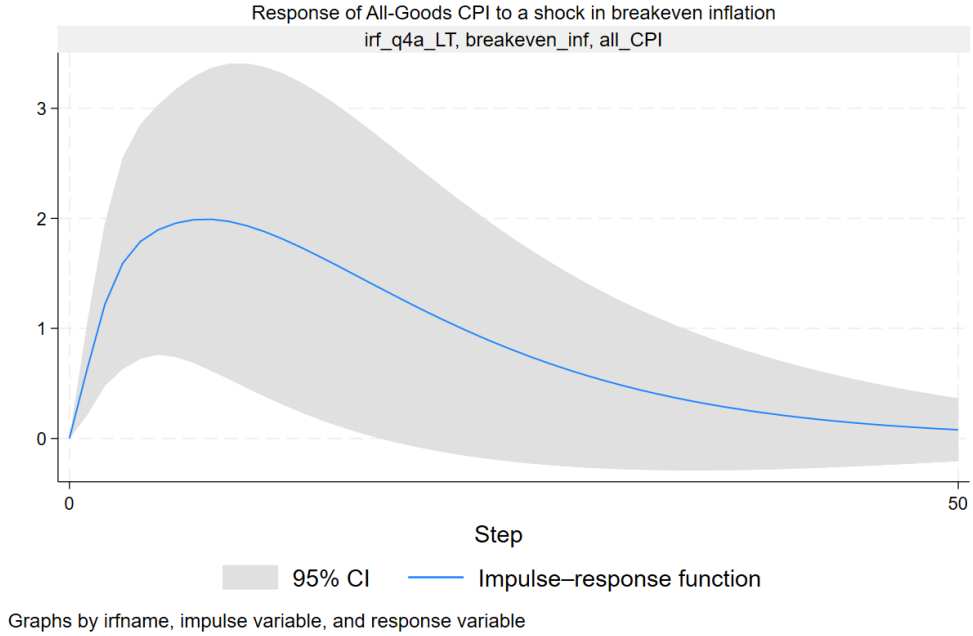

Figure 4a-1: IRF of All-Goods CPI to a shock in breakeven inflation

Question 4b:

Separating the effects of a price shock and an expectations shock can be done through imposing restrictions using a SVAR, explicitly defining relationships between variables transparently through the A matrix. I impose that πbt reacts contemporaneously to πt and will only have a lagged effect on πt. Ordering πt first is justified by sticky prices, where frictions like menu costs prevent firms from adjusting instantly, resulting in πbt affecting πt only with a lag. Ordering πbt second means it reacts contemporaneously to current prices, consistent with rational expectations, where agents update beliefs upon observing new prices.

Figure 4a-1 compares the orthogonalized and naïve IRF: the difference directly shows how the naïve estimate overstates the pure expectations channel. In the SVAR, the pure expectations shock only accounts for 16% of the total variance in inflation after 12 months, whereas a pure price shock accounts for around 83%.

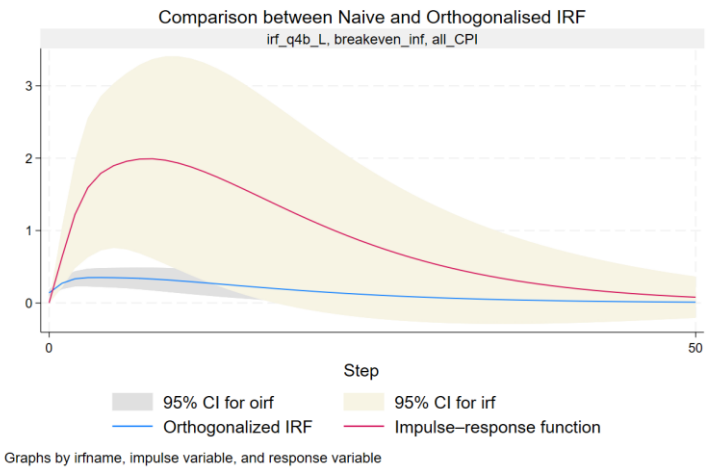

Figure 4a – 1: Naïve vs Orthogonalized IRF comparison

Question 5:

ADF tests confirmed the non-stationary of πt, bt, and rt. A Johansen test (restricted constant) indicated a rank of 2, but the VECM showed a spurious relationship between πt and rt.

The significant relationship between bt, and rt, implies the stationarity of breakeven inflation, consistent with anchored expectations. Normalising the coefficient for rt allows the relationship to be interpreted as the Fisher equation. The coefficient of bt with confidence interval of (-0.93, -0.77) suggests nominal yields move slightly less than one-for-one with real yields in the long run. This implies a time-varying risk premium, which could be a result of outlier periods like 2008 and 2021.

Figure 5-2 illustrates that each period, bt corrects for disequilibrium by 4% each period, whilst rt corrects it by 12%, but insignificantly. Excluding rt and πt from the cointegrating vector found that rt was significant to the relationship (LR test statistic of 9.994), whilst πt was not (test statistic of 0.774), reinforcing the interpretation that bond markets anchor inflation expectations independently of realised inflation.

Variable	Coefficient	Standard error	p-value	Confidence interval
π_t	0.0030	0.0015	0.039	(0.0002, 0.0058)
r_t	-0.0025	0.0006	0.000	(-0.0037, -0.0012)
b_t	-0.0016	0.0007	0.017	(-0.0031, -0.0003)

Figure 5-2: Adjustment coefficients